

How quantum is the advantage? A quantum-kernel result, audited

1 Reported

A quantum kernel beats its classical random-feature surrogate on NSL-KDD

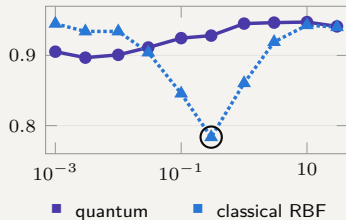
+0.134

ROC-AUC

Significant after false-discovery-rate *and* Holm correction

2 Audited

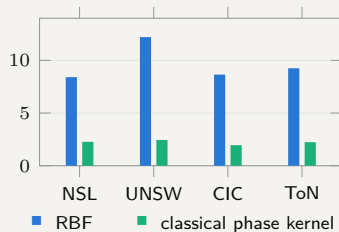
Same data, exact kernels, equal tuning. Test AUC by bandwidth:



Gap set by the *classical* grid:
 $\gamma \leq 3$: +0.167 $\gamma \leq 30$: +0.007

3 Answer

Geometric difference g to the IQP kernel, four datasets:



No quantum state needed.
Fix: leave-attack-types-out CV